

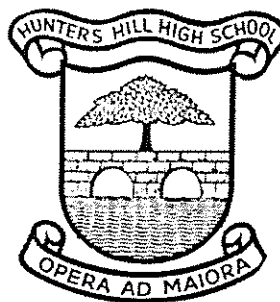
Student Name: _____

HUNTERS HILL HIGH SCHOOL

EXTENSION 2

MATHEMATICS

HSC TRIAL 2017



Hunters Hill
High School

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Board-approved calculators may be used
- A Reference Sheet is provided
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

Total marks - 100

Section I

10 marks

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II

90 marks

- Attempt Questions 11-16
- Allow about 2 hour 45 minutes for this section

Section I

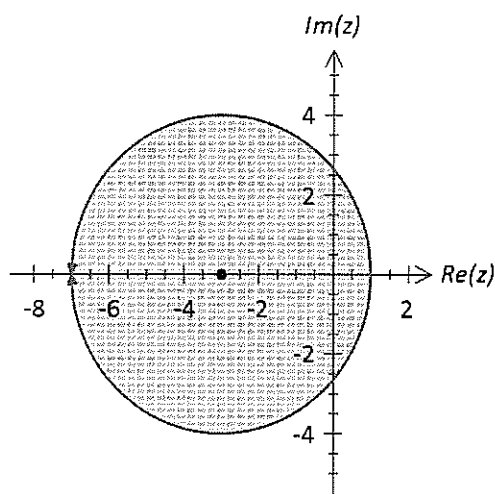
10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10.

1



Which of the following inequalities describes the region above?

(A) $|z - 3| \leq 2$

(B) $|z + 3| \leq 2$

(C) $|z - 3| \leq 4$

(D) $|z + 3| \leq 4$

2

What is the value of i^{2017} ?

(A) i

(B) 1

(C) $-i$

(D) -1

- 3 Multiplying a non-zero complex number by $\frac{1+i}{1-i}$ results in a rotation about the origin on an Argand diagram. What is the rotation?

- (A) Clockwise by $\frac{\pi}{4}$
- (B) Clockwise by $\frac{\pi}{2}$
- (C) Anticlockwise by $\frac{\pi}{4}$
- (D) Anticlockwise by $\frac{\pi}{2}$

- 4 What is the eccentricity of $\frac{x^2}{9} + \frac{y^2}{16} = 1$?

- (A) $\frac{7}{16}$
- (B) $\frac{\sqrt{7}}{4}$
- (C) $\frac{9}{16}$
- (D) $\frac{7}{9}$

- 5 The gradient of the curve $x^2y - xy^2 + 6 = 0$ at the point $P(2,3)$ is equal to:

- (A) -5 (B) $\frac{3}{8}$
- (C) $\frac{9}{8}$ (D) 1

6 The directrices of the hyperbola $\frac{y^2}{9} - \frac{x^2}{16} = 1$ are

(A) $x = \pm \frac{9}{5}$ (B) $y = \pm \frac{9}{5}$

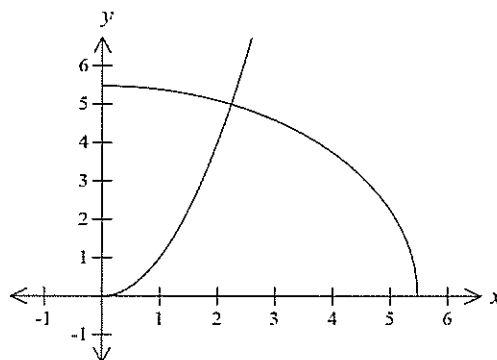
(C) $y = \pm 5$ (D) $x = \pm 5$

7 Which of the following is the correct expression for $\int \frac{1}{\sqrt{8+2x-x^2}} dx$?

(A) $\sin^{-1} \frac{x-1}{9} + C$ (B) $\sin^{-1} \frac{x-1}{3} + C$

(C) $\sin^{-1} \frac{x+1}{3} + C$ (D) $\sin^{-1} \frac{x+1}{9} + C$

8 What is the volume of the solid formed when the region bounded by the curves $y = x^2$, $y = \sqrt{30-x^2}$ and the y -axis is rotated about the y -axis? Use the method of slicing.



What is the correct expression for volume of this solid using the method of cylindrical shells?

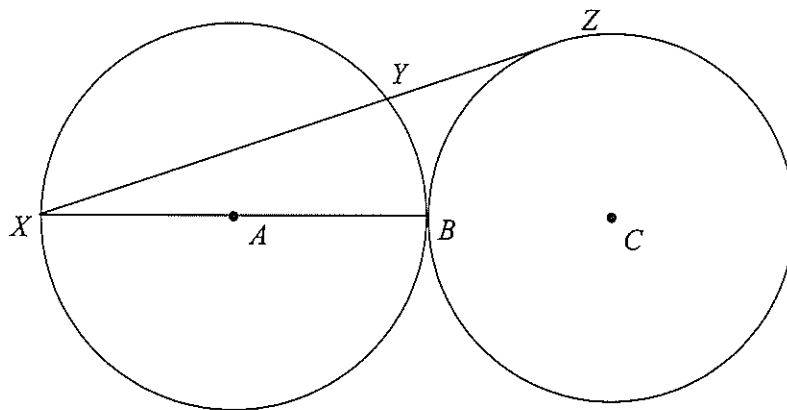
(A) $V = \int_0^{\sqrt{5}} 2\pi \left(x^2 - \sqrt{30-x^2} \right) dx$

(B) $V = \int_0^{\sqrt{5}} 2\pi x \left(x^2 - \sqrt{30-x^2} \right) dx$

(C) $V = \int_0^{\sqrt{5}} 2\pi \left(\sqrt{30-x^2} - x^2 \right) dx$

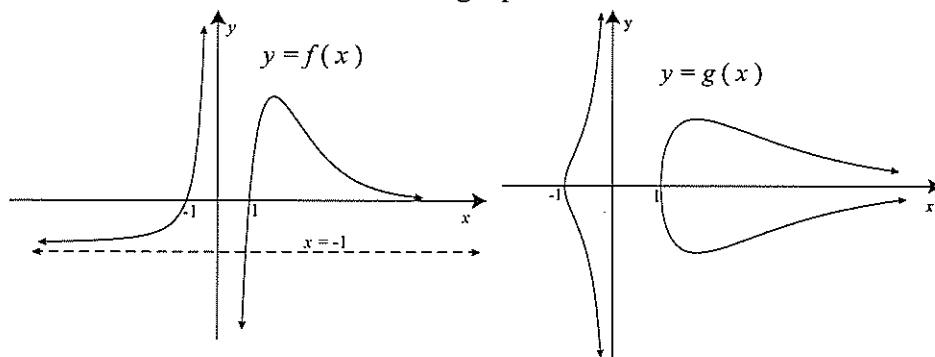
(D) $V = \int_0^{\sqrt{5}} 2\pi x \left(\sqrt{30-x^2} - x^2 \right) dx$

- 9 Two equal circles touch externally at B . XB is a diameter of one circle. XZ is the tangent from X to the other circle and cuts the first circle at Y .



Which is the correct expression that relates XZ to XY ?

- (A) $3XZ = 4XY$
 (B) $XZ = 2XY$
 (C) $2XZ = 3XY$
 (D) $2XZ = 5XY$
- 10 What statement is true for these graphs?



- (A) $|f(x)| = g(x)$
 (B) $g(x) = \ln[f(x)]$
 (C) $f(x) = \pm \ln|g(x)|$
 (D) $g(x) = \pm \sqrt{f(x)}$

Section II

90 marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11-16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) Find

2

(i) $\int \frac{\cos \theta}{\sin^5 \theta} d\theta$

2

(ii) $\int \frac{dx}{x^2 + 2x + 2}$

(b)

Evaluate $\int_0^1 \frac{2x+1}{x^2+1} dx$

3

(c) Evaluate the following definite integrals:

(i) $\int_0^1 \cos^{-1} x \, dx$

2

(ii) $\int_1^2 x(\ln x)^2 \, dx$

3

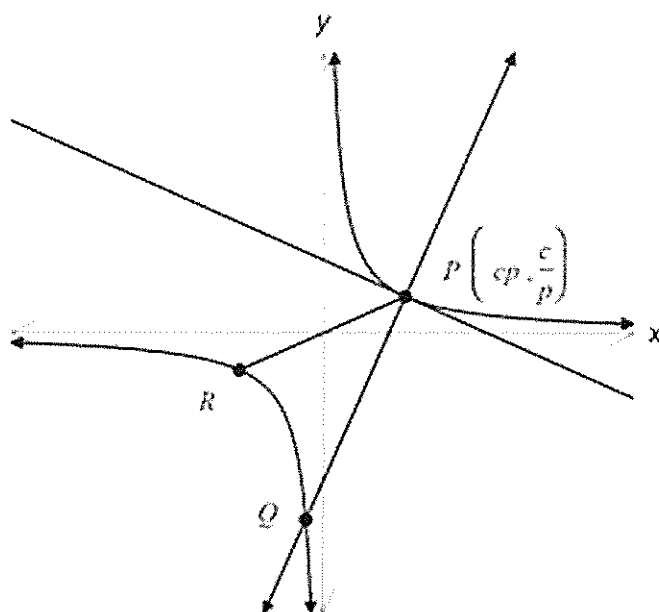
(d)

Use the substitution $x = \frac{1}{u}$ to evaluate $\int_{\frac{1}{e}}^e \frac{\log_e x}{(1+x)^2} dx$

3

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) PQ is a variable chord of the rectangular hyperbola $xy = c^2$.

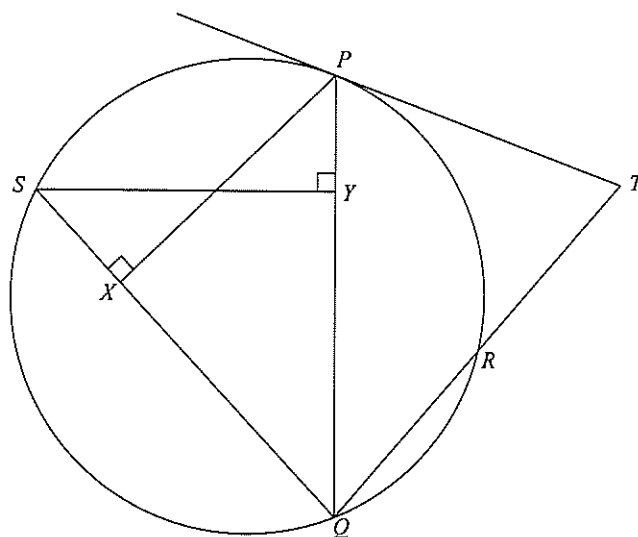


- (i) Show that the equation of the normal at the point $P\left(cp, \frac{c}{p}\right)$ on the rectangular hyperbola $xy = c^2$ is given by $p^3x - py = c(p^4 - 1)$ 2
- (ii) Prove that the normal cuts the hyperbola again at the point $Q\left(\frac{-c}{p^3}, -cp^3\right)$ 3
- (iii) If R is the opposite end of the diameter of the hyperbola through P, show that PR is perpendicular to RQ . 2

Question 12 continued on next page.

Question 12 (continued)

- (b) In the diagram below, TP is the tangent of the circle at P , and TQ is a secant cutting the circle at R . SQ is a chord of the circle such that PX and SY are perpendicular to SQ and PQ , respectively.



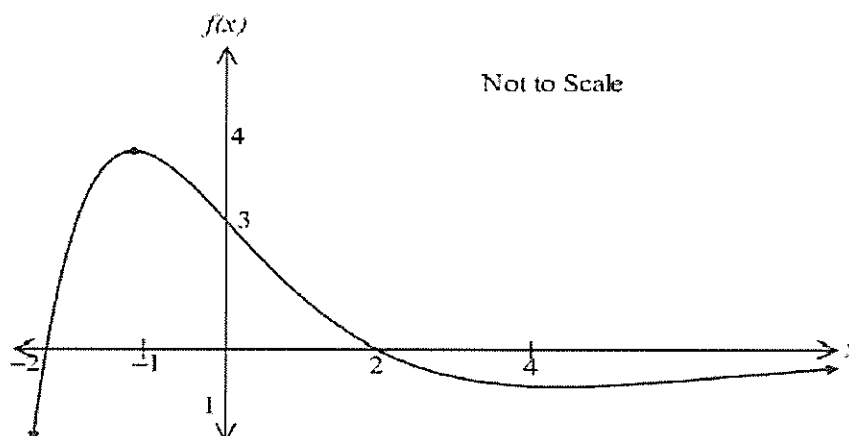
- (i) Prove that $\angle TRP = \angle TPQ$ 3
- (ii) Explain why $SPYX$ is a cyclic quadrilateral and state the diameter of the circle. 1
- (iii) Prove $\angle PYX = \angle PRQ$ 2
- (c) If P and Q represent the complex numbers z and w , where 2

$$w = \frac{1}{z-2} + \frac{3}{2},$$

find the Cartesian equation of the locus of Q as P moves on the circle $|z-2|=3$.

Question 13 (15 marks) Use a SEPARATE writing booklet.

(a)



The diagram shows the graph of the function, which has a horizontal asymptote at $y = 0$. On separate diagrams sketch the following graphs, showing clearly any intercepts on the coordinate axes and the equations of any asymptotes:

(i) $y = f(x^2)$ 2

(ii) $y = \frac{1}{f(x)}$ 2

(iii) $y = \ln(f(x))$ 2

(b) Sketch $|y| = x^2 - 4x$, showing all important features. 4

(c) If $z_1 = 1 - i$, $z_2 = 2z_1$ and $z_3 = -2iz_1$ clearly on an Argand diagram the points represented by

(iii) z_1, z_2 and z_3 3

(iv) $z_3 - z_2$ 2

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) The equation $x^4 + 2x^3 - 7x^2 - 20x - 12 = 0$ has a double root. Find this root and hence solve this equation. 3

- (b) The equation $x^3 + px + 5 = 0$ has roots α , β and γ .

- (i) Find in terms of p , $\alpha^2 + \beta^2 + \gamma^2$ 2

- (ii) Show that $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 = p^2$ 2

- (iii) Find the cubic polynomial with integer coefficients, whose roots are

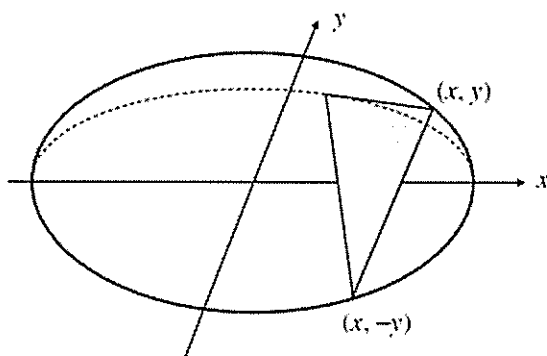
$$\frac{\alpha}{\beta\gamma}, \frac{\beta}{\gamma\alpha}, \frac{\gamma}{\alpha\beta}.$$
 2

- (c) Solve the equation $x^4 + 2x^3 + x^2 - 1 = 0$, given that one root is 3

$$-\frac{1}{2} + i\frac{\sqrt{3}}{2}$$

- (d) The base of a solid is in the shape of an ellipse with equation $\frac{x^2}{9} + \frac{y^2}{4} = 1$.

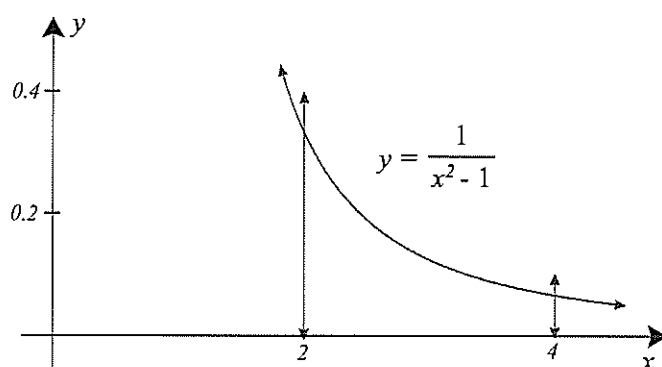
Sections parallel to the y -axis are equilateral triangles, with one side sitting in the base of the solid, as shown in the diagram.



Find the volume of this solid. 3

Question 15 (15 marks) Use a SEPARATE writing booklet.

- (a) The region bounded by the curve $y = \frac{1}{x^2 - 1}$ and the x -axis between $x = 2$ and $x = 4$ is rotated through one revolution about the line $x = 2$.

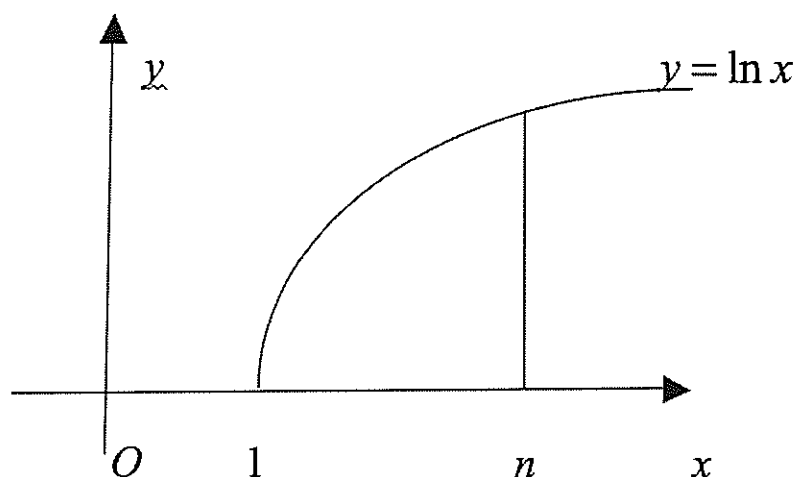


- (i) Use the method of cylindrical shells to show that the volume, V , of the solid formed is given by $V = 2\pi \int_2^4 \frac{x-2}{x^2-1} dx$. 2
- (ii) Hence find the exact value of V in simplest form. 3
- (b) If $a > 0$, $b > 0$, $c > 0$ and $a + b + c = 1$, (use in part (ii) only)
- (i) show that $(a+b)(b+c)(c+a) \geq 8abc$ 3
- (ii) hence, or otherwise, prove that 2
- $$(1-a)(1-b)(1-c) \geq 8abc$$
- (c) (i) Prove that $\int_0^a F(x) dx = \int_0^a F(a-x) dx$. 2
- (ii) Hence, or otherwise, evaluate $\int_0^{\frac{\pi}{4}} \log_e(1 + \tan \theta) d\theta$ 3

Question 16 (15 marks) Use a SEPARATE writing booklet.

- (a) Twelve different books are made into four parcels of three each. How many different sets of parcels could be made? 3

(b)



- (i) Use the trapezoidal rule with n function values to approximate $\int_1^n \ln x \, dx$. 2

- (ii) Show that $\frac{d}{dx}(x \ln x - x) = \ln x$ and hence find the exact value of $\int_1^n \ln x \, dx$. 2

- (i) Deduce that $\ln n! < \left(n + \frac{1}{2}\right) \ln n - n + 1$. 2

(b)
$$I_n = \int_1^e (1 - \ln x)^n \, dx, \quad n = 1, 2, 3, \dots$$

- (i) Show $I_n = -1 + nI_{n-1}$, $n = 1, 2, 3, \dots$ 2

- (ii) Hence evaluate $\int_1^e (1 - \ln x)^4 \, dx$. 2

- (iii) Show that $\frac{I_n}{n!} = e - \sum_{r=0}^n \frac{1}{r!}$, $n = 1, 2, 3, \dots$ 2

END OF EXAMINATION

MXX TRIAL 2018

SECTION I.

1. D

2. A

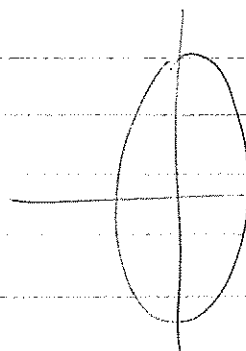
$$2017 = 4 \times 504 + 1.$$

3. D

$$\frac{1+i}{1-i} \times \frac{1+i}{1+i} = \frac{2i}{1+1} = i$$

4. B

$$e^2 = 1 - \frac{b^2}{a^2} = 1 - \frac{16}{9} \cdot \frac{9}{16} = \frac{7}{6}$$



5. B

$$2xy + x^2 \frac{dy}{dx} - y^2 - 2xy \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} (x^2 - 2xy) = y^2 - 2xy$$

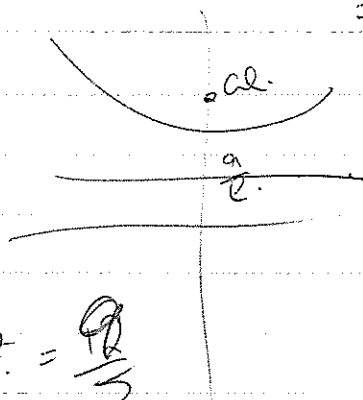
$$\frac{dy}{dx} = \frac{3^2 - 2(2)(3)}{2^2 - 2(2)(3)} = \frac{9 - 12}{4 - 12} = \frac{-3}{-8}$$

6. B

$$e^2 = 1 + \frac{46}{16} = \frac{25}{4}$$

$$e = \frac{5}{2}$$

$$\frac{3 \cdot 3}{5} = \frac{9}{5}$$

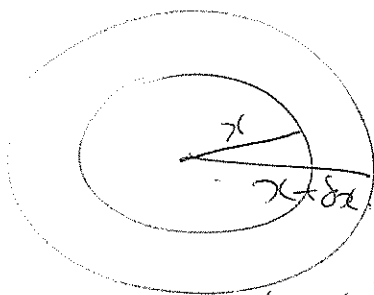


7. B.

$$\frac{1}{\sqrt{8+2x-x^2}} = \frac{1}{\sqrt{9-1+2x-x^2}}$$

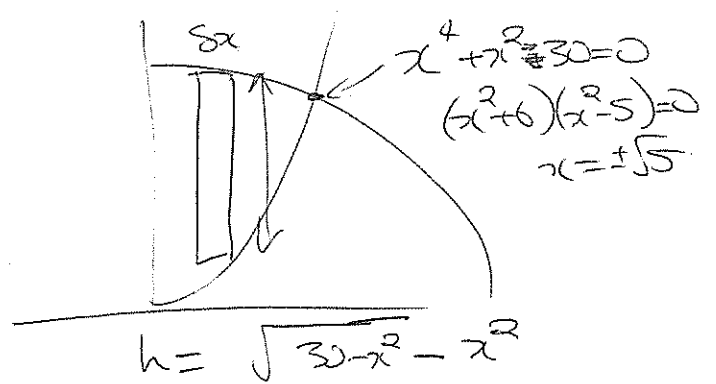
$$= \frac{1}{\sqrt{9-(x-1)^2}}$$

8. D



$$\pi((x+\delta x)^2 - x^2)$$

$$= \pi(2x\delta x)$$



9. C

$$\frac{x^4}{x^2} = \frac{2}{3}$$

10. D.

SECTION II

Question II

a) i $I = \int \frac{\cos \theta}{\sin^5 \theta} d\theta$

let $u = \sin \theta$.

$du = \cos \theta d\theta$.

$$I = \int u^{-5} du$$

$$= \frac{u^{-4}}{-4} + C$$

$$= -\frac{1}{4\sin^4 \theta} + C$$

ii. $I = \int \frac{dx}{x^2 + 2x + 2}$

$$= \int \frac{dx}{(x+1)^2 + 1}$$

$$= \tan^{-1}(x+1) + C$$

b) $\int_0^1 \frac{2x+1}{x^2+1} dx = \int_0^1 \left(\frac{2x}{x^2+1} + \frac{1}{x^2+1} \right) dx$

$$= \left[\ln(x^2+1) + \tan^{-1} x \right]_0^1$$

$$= \ln(1^2+1) + \tan^{-1} 1 - \ln(0^2+1) - \tan^{-1} 0$$

$$= \ln 2 + \frac{\pi}{4}$$

c) i) $I = \int_0^1 \cos^{-1} x \, dx$

let $u = \cos^{-1} x$ $du = \frac{-1}{\sqrt{1-x^2}} dx$
 $dx = -\sqrt{1-x^2} du$

$v = x$

$= \left[x \cos^{-1} x \right]_0^1 + \int_0^1 \frac{x}{\sqrt{1-x^2}} dx$

let $u = 1-x^2$

$du = -2x dx$

$I = (1 \cos^{-1} 1 - 0 \cos^{-1} 0) - \frac{1}{2} \int_1^0 \frac{du}{\sqrt{u}}$

$= \frac{1}{2} \left[\frac{2}{1} u^{\frac{1}{2}} \right]_0^1$

$= 1$

ii) $\int_1^2 x (\ln x)^2 dx$

let ~~$x = e^u$~~ $u = \ln x$
 $du = \frac{dx}{x}$

~~$e^u du = dx$~~

~~$I = \int_0^{\ln 2} e^u u^2 du$~~

let $u = (\ln x)^2$ ~~$du = \frac{2 \ln x}{x} dx$~~

~~$du = x dx$ $v = \frac{x^2}{2}$~~

$I = \left[\frac{x^2}{2} (\ln x)^2 \right]_1^2 - \int_1^2 \frac{2 \ln x}{x} \frac{x^2}{2} dx$

$= \left[\frac{2^2 (\ln 2)^2}{2} - \frac{1^2 (\ln 1)^2}{2} \right] - \int_1^2 x \ln x dx$

$u = \ln x$ $du = \frac{dx}{x}$
 $du = x dx$ $v = \frac{x^2}{2}$

$= 2(\ln 2)^2 - \left[\frac{x^2}{2} \ln x \right]_1^2 + \int_1^2 \frac{x^2}{2} \frac{dx}{x}$

$= 2(\ln 2)^2 - \frac{2^2 \ln 2}{2} + \frac{1^2 \ln 1}{2} + \left[\frac{x^2}{2} \right]_1^2 = 2(\ln 2)^2 - 2 \ln 2 + \frac{3}{4}$

$$d) \int_{\frac{1}{e}}^e \frac{\log_e x}{(1+x)^2} dx$$

$$\text{let } x = \frac{1}{u}$$

$$\begin{aligned} \log_e x &= \log_e \frac{1}{u} \\ &= -\log_e u \end{aligned}$$

$$x=e, \quad u=\frac{1}{e}$$

$$x=\frac{1}{e}, \quad u=e$$

$$I = \int_e^{\frac{1}{e}} \frac{-\log_e u}{(1+\frac{1}{u})^2} \left(-\frac{1}{u^2}\right) du$$

$$= + \int_e^{\frac{1}{e}} \frac{\log_e u}{\cancel{u^2} \frac{1}{u^2} (u+1)^2} du$$

$$= - \int_{\frac{1}{e}}^e \frac{\log_e u}{(1+u)^2} du = -I$$

$$\therefore 2I = 0$$

$$I = 0$$

Question 12.

- a) i)

$$xy = c^2.$$

$$y = \frac{c^2}{x}$$

$$\frac{dy}{dx} = -\frac{c^2}{x^2}$$

at P

$$m_T = \frac{-c^2}{(cp)^2}$$

$$\Rightarrow m_N = p^2.$$

$$= -\frac{1}{p^2}$$

by point-gradient

$$y - \frac{c}{p} = p^2(x - cp)$$

$$py - c = p^3x - cp^4.$$

$$\therefore p^3x - py = cp^4 - c.$$

so $p^3x - py = c(p^4 - 1)$ is the equation of the normal.

ii) Q $(-\frac{c}{p^3}, -cp^3)$

Test in normal

$$\text{LHS} = p^3\left(-\frac{c}{p^3}\right) - p(-cp^3)$$

$$= -c + cp^4$$

$$= c(p^4 - 1)$$

$$= \text{RHS}$$

Test in hyperbola

$$\text{LHS} = \left(-\frac{c}{p^3}\right)(-cp^3)$$

$$= c^2$$

$$= \text{RHS}$$

as Q satisfies both, it is intersection of normal and hyperbola.

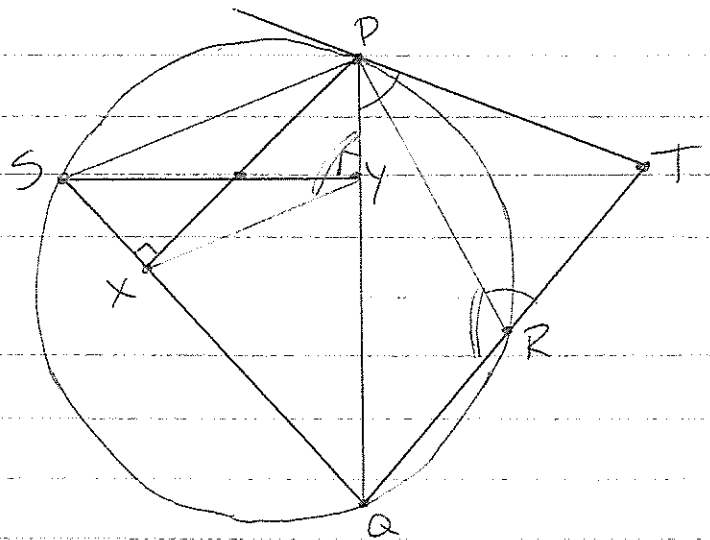
iii) as $PQ \left(CP, \frac{c}{P} \right)$,

R is $\left(-CP, -\frac{c}{P} \right)$

$$\begin{aligned}
 M_{PR} \cdot M_{QR} &= \frac{\frac{c}{P} - \left(-\frac{c}{P} \right)}{CP - (-CP)} \cdot \frac{-CP^3 - \left(-\frac{c}{P} \right)}{-\frac{c}{P^3} - (-CP)} \\
 &= \frac{\cancel{2c}}{P} \left(\frac{1}{\cancel{2CP}} \right) \cdot \frac{-CP^4 + c^{-1} \cdot \frac{1}{P}}{-c + CP^4 \cdot \frac{1}{P^3}} \\
 &= \frac{1}{P^2} \cdot -P^2 \\
 &= -1
 \end{aligned}$$

$\therefore PR \perp QR$

b)



i) In $\triangle TPR$ and $\triangle TQP$
 $\angle T = \angle T$ (common)
 $\angle TPR = \angle TQP$ (angle in alternate segment)
 $\therefore \triangle TPR \parallel \triangle TQP$ (equiangular)
 and $\angle TRP = \angle TPQ$ (corresponding angles of similar triangles)

ii) as $\angle SXP = \angle SYP = 90^\circ$,
 S, P, Y, X are concyclic,
 Hence, $SPYX$ is a cyclic quadrilateral
 with diameter SP .

iii) $\angle PYX = 180^\circ - \angle PSX$ (opposite angles of cyclic quadrilateral $SPYX$ are supplementary)
 Similarly in $SPRQ$.

$$\angle PRQ = 180^\circ - \angle PSX$$

$$\text{Hence } \angle PRQ = \angle PYX.$$

c)

$$\omega = \frac{1}{z-2} + \frac{3}{2} \quad \frac{1}{z-2} = \frac{2\omega-3}{2}$$

$$|z-2| = \left| \frac{2}{2\omega-3} \right|$$

$$= \frac{1}{z-2} \cdot \frac{\overline{(z-2)}}{\overline{(z-2)}} + \frac{3}{2}$$

$$= \frac{x-2-iy}{|z-2|} + \frac{3}{2}$$

$$3 = \left| \frac{2}{2x-3+2iy} \right|$$

$$\frac{3}{2} = \left| \frac{2x-3-2iy}{(2x-3)^2 + (2y)^2} \right|$$

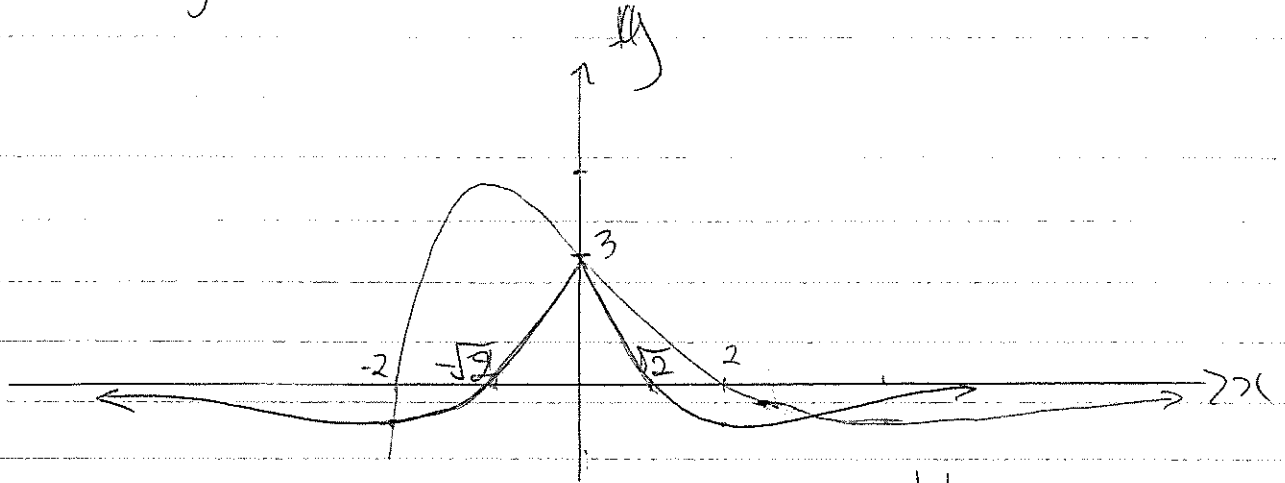
$$\frac{9}{4} = \frac{(2x-3)^2 + (2y)^2}{((2x-3)^2 + (2y)^2)^2}$$

$$(2x-3)^2 + (2y)^2 = \frac{4}{9}$$

$$\left(x - \frac{3}{2}\right)^2 + y^2 = \frac{1}{9}$$

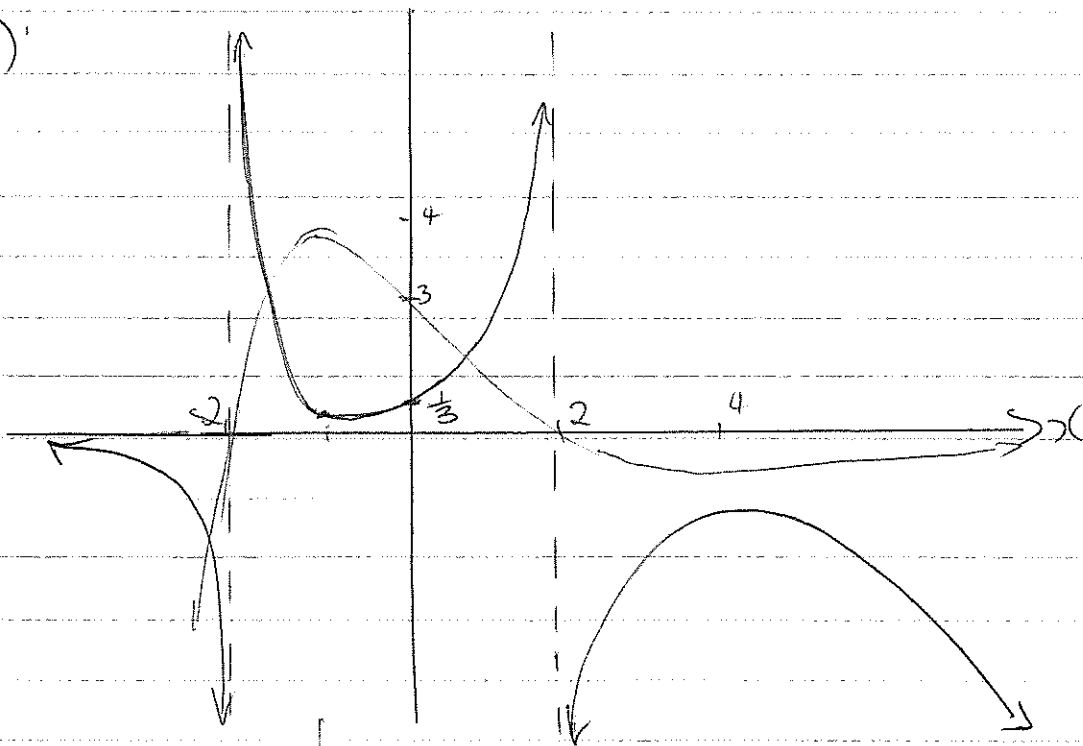
Question B

a) $y = f(x^2)$

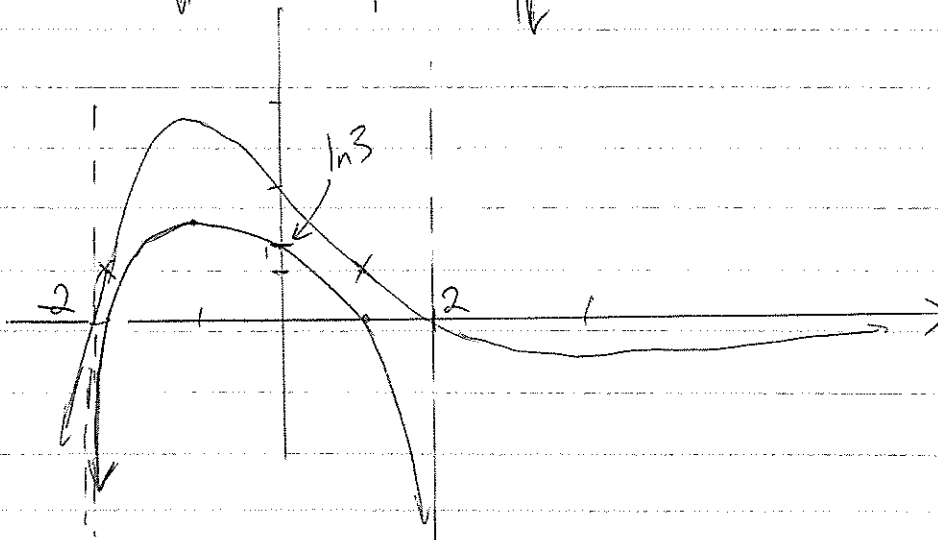


asymptote: $y = 0$

b)



c



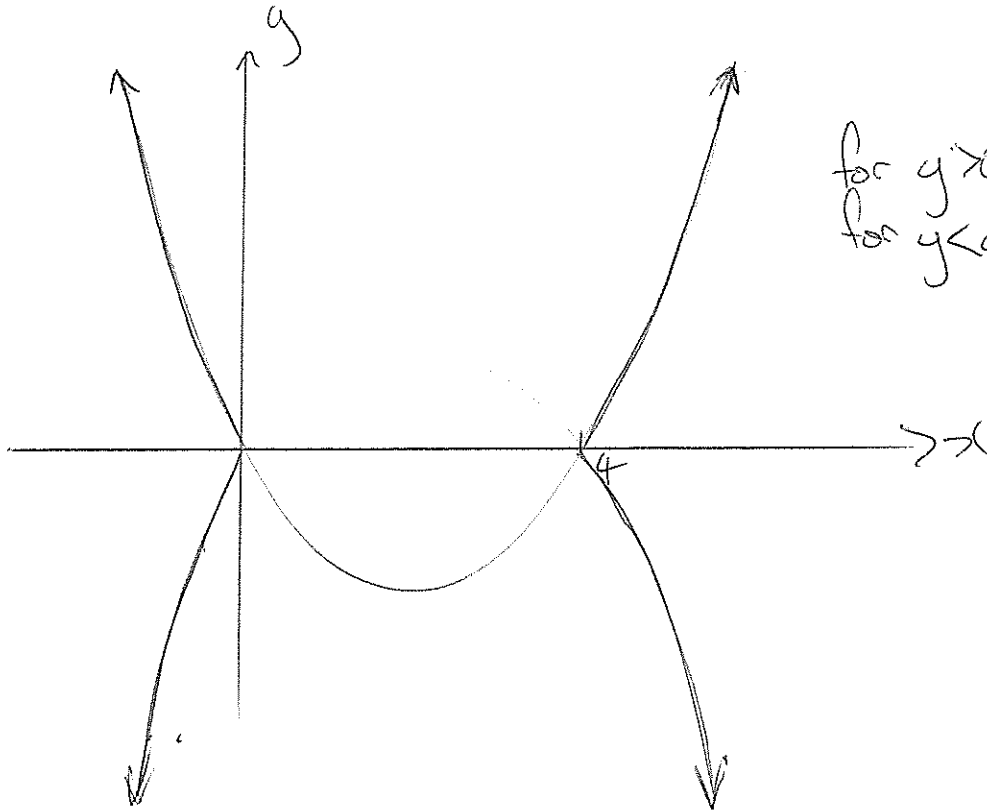
b).

$$y = x^2 - 4x$$

$$= x(x-4)$$

$$x\text{-ints } x=0, 4,$$

$$y\text{-int } 0.$$

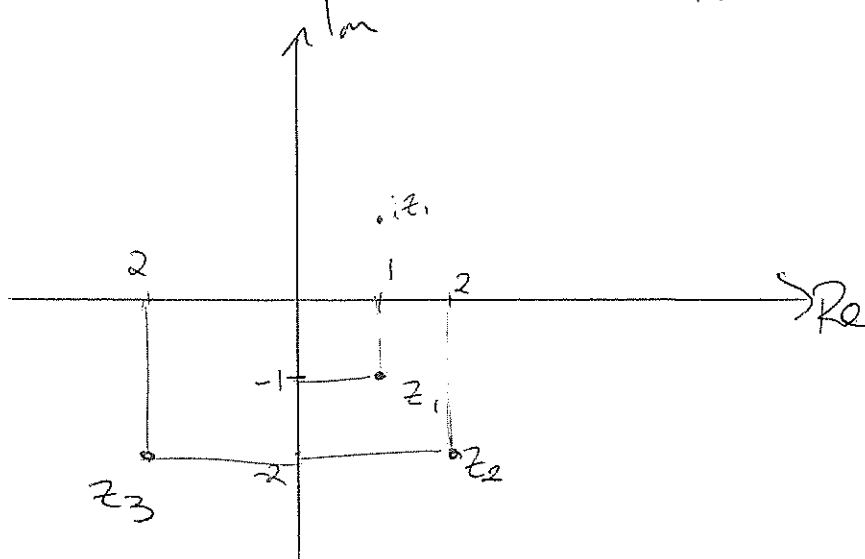


$$\text{for } y > 0, |y| = x^2 - 4x$$

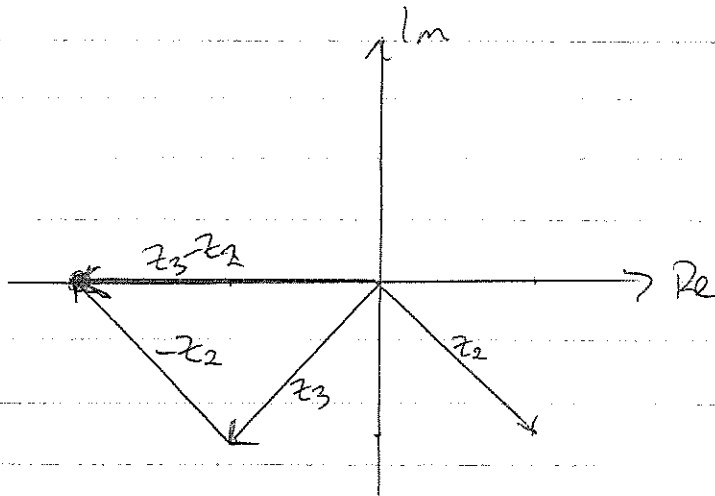
$$\text{for } y < 0, |y| = -(x^2 - 4x)$$

c) $z_1 = 1 - i, z_2 = 2z_1, z_3 = -2iz_1.$

iii)



10)



Question 14

$$a) \quad P(x) = x^4 + 2x^3 - 7x^2 - 20x - 12$$

$$P'(x) = 4x^3 + 6x^2 - 14x - 20$$

test for roots in $P'(x)$

$$P'(2) = 4(2)^3 + 6(2)^2 - 14(2) - 20$$

$$= 8$$

$$P'(-2) = 4(-2)^3 + 6(-2)^2 - 14(-2) - 20$$

$$= 0$$

test $x = -2$ in $P(x)$

$$P(-2) = (-2)^4 + 2(-2)^3 - 7(-2)^2 - 20(-2) - 12$$

$$= 0$$

$\therefore x = -2$ is the double root as it satisfies $P(x) = 0$ and $P'(x) = 0$

$$\begin{array}{r}
 x^2 - 2x - 3 \\
 \hline
 x^2 + 4x + 4 \quad) \quad x^4 + 2x^3 - 7x^2 - 20x - 12 \\
 \underline{x^4 + 4x^3 + 4x^2} \\
 -2x^3 - 11x^2 - 20x \\
 \underline{-2x^3 - 8x^2 - 8x} \\
 -3x^2 - 12x - 12 \\
 \underline{-3x^2 - 12x - 12} \\
 0
 \end{array}$$

$$\therefore x^4 + 2x^3 - 7x^2 - 20x - 12 = (x+2)^2 (x^2 - 2x - 3) \\ = (x+2)^2 (x-3)(x+1)$$

$$\therefore (x+2)^2 (x-3)(x+1) = 0$$

$x = -2, -1, 3$ is the solution.

b) $x^3 + px + 5 = 0$ roots are α, β, γ .

i) $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) \\ = \left(-\frac{0}{1}\right)^2 - 2\left(\frac{p}{1}\right) \\ = -2p$

ii) $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 = (\alpha\beta)^2 + (\beta\gamma)^2 + (\gamma\alpha)^2 \\ = (\alpha\beta + \beta\gamma + \gamma\alpha)^2 - 2(\alpha\beta \cdot \beta\gamma + \beta\gamma \cdot \gamma\alpha + \gamma\alpha \cdot \alpha\beta) \\ = (\alpha\beta + \beta\gamma + \gamma\alpha)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma) \\ = (p)^2 - 2\left(-\frac{5}{1}\right)(0) \\ = p^2$

iii) roots $\frac{\alpha}{\beta\gamma}, \frac{\beta}{\gamma\alpha}, \frac{\gamma}{\alpha\beta}$.

sum of roots: $\frac{\alpha + \beta + \gamma}{\beta\gamma \cdot \gamma\alpha \cdot \alpha\beta} = \frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha\beta\gamma} \\ = \frac{-2p}{-5} \\ = \frac{2p}{5}$

sum of roots: $\frac{\alpha}{\beta\gamma} + \frac{\beta}{\gamma\alpha} + \frac{\gamma}{\alpha\beta}$

$$= \frac{1}{\gamma^2} + \frac{1}{\alpha^2} + \frac{1}{\beta^2}$$

$$= \frac{\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2}{\alpha^2\beta^2\gamma^2}$$

$$= \frac{p^2}{(-5)^2}$$

$$= \frac{p^2}{25}$$

product of roots: $\frac{\alpha}{\beta\gamma} \cdot \frac{\beta}{\gamma\alpha} \cdot \frac{\gamma}{\alpha\beta} = \frac{1}{\alpha\beta\gamma}$

$$= \frac{1}{(-5)}$$

$$= -\frac{1}{5}$$

polynomial is

$$x^3 + \left(\frac{-2p}{5}\right)x^2 + \frac{p^2}{25}x + \left(\frac{1}{5}\right) = 0.$$

$$\therefore -25x^3 - 10px^2 + p^2x + 5 = 0$$

is the polynomial with roots $\frac{\alpha}{\beta\gamma}, \frac{\beta}{\gamma\alpha}, \frac{\gamma}{\alpha\beta}$

$$c) \quad x^4 + 2x^3 + x^2 - 1 = 0$$

as $-\frac{1}{2} + i\frac{\sqrt{3}}{2}$ is a root, so is $-\frac{1}{2} - i\frac{\sqrt{3}}{2}$.

$$\begin{aligned} & \left(x - \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)\right) \left(x - \left(-\frac{1}{2} - i\frac{\sqrt{3}}{2}\right)\right) \\ &= \left(\left(x + \frac{1}{2}\right) - i\frac{\sqrt{3}}{2}\right) \left(\left(x + \frac{1}{2}\right) + i\frac{\sqrt{3}}{2}\right) \\ &= \left(x + \frac{1}{2}\right)^2 + \frac{3}{4} \\ &= x^2 + x + \frac{1}{4} + \frac{3}{4} \end{aligned}$$

$= x^2 + x + 1$ is a factor of polynomial

$$\begin{array}{r} x^2 + x + 1 \overline{) x^4 + 2x^3 + x^2 - 1} \\ \underline{x^4 + x^3 + x^2} \\ x^3 \\ \underline{x^3 + x^2 + x} \\ -x^2 - x - 1 \\ \underline{-x^2 - x - 1} \\ 0 \end{array}$$

$$\therefore (x^2 + x + 1)(x^2 + x - 1) = 0$$

$$x^2 + x - 1 = 0$$

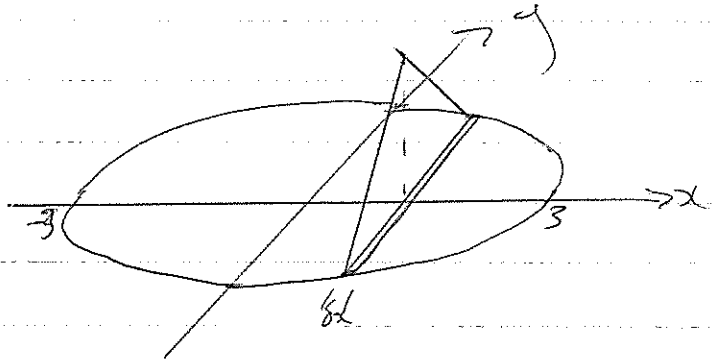
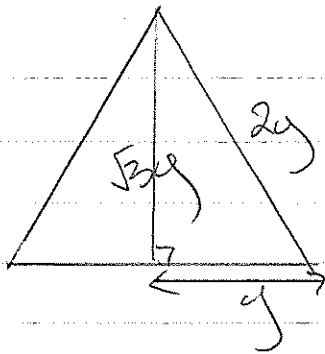
$$x = \frac{-1 \pm \sqrt{1^2 - 4(1)(-1)}}{2(1)}$$

$$= \frac{-1 \pm \sqrt{5}}{2}$$

$$\therefore x = -\frac{1}{2} \pm i\frac{\sqrt{3}}{2}, -\frac{1}{2} \pm \frac{\sqrt{5}}{2}$$

d)

$$\frac{x^2}{9} + \frac{y^2}{4} = 1$$



$$\begin{aligned} A &= \frac{1}{2} b h \\ &= \frac{1}{2} \times 2y \times \sqrt{3}y \\ &= \sqrt{3} y^2 \end{aligned}$$

$$\delta V = \sqrt{3} y^2 \delta x$$

$$= \sqrt{3} \frac{4}{9} (9 - x^2) \delta x$$

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=-3}^3 \frac{4\sqrt{3}}{9} (9 - x^2) \delta x$$

$$= \frac{4\sqrt{3}}{9} \int_{-3}^3 (9 - x^2) dx$$

$$= \frac{4\sqrt{3}}{9} 2 \int_0^3 (9 - x^2) dx$$

$$f(x) = 9 - x^2 \text{ is even}$$

$$= \frac{8\sqrt{3}}{9} \left[9x - \frac{x^3}{3} \right]_0^3$$

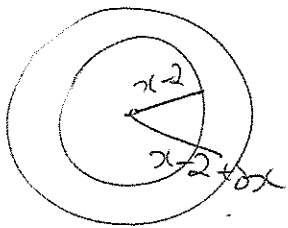
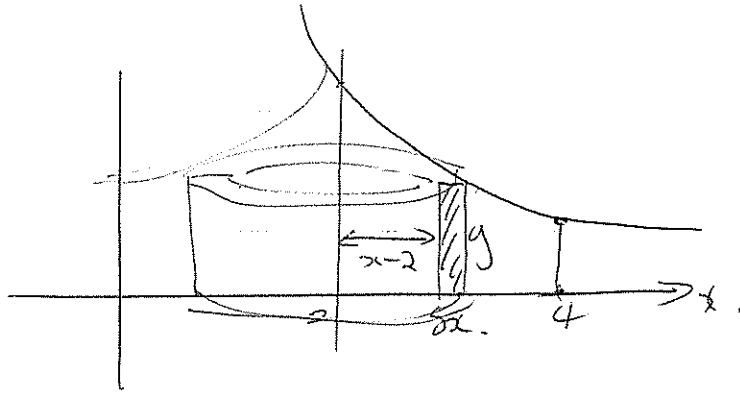
$$= \frac{8\sqrt{3}}{9} \left(9(3) - \frac{3^3}{3} - 9(0) + \frac{0^3}{3} \right)$$

$$= \frac{8\sqrt{3}}{9} (27 - 9)$$

$$= 16\sqrt{3} \text{ units}^3$$

Question 15.

a) i)



$$\delta A = \pi \left[(x-2+\delta x)^2 - (x-2)^2 \right]$$

$$= \pi \left[(x-2)^2 + 2(x-2)\delta x + \delta x^2 - (x-2)^2 \right]$$

$$= 2\pi (x-2) \delta x$$

$$\delta V = 2\pi (x-2) \delta x \cdot y$$

$$= 2\pi (x-2) \delta x \cdot \frac{1}{x^2-1}$$

$$= 2\pi \frac{x-2}{x^2-1} \cdot \delta x$$

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=2}^4 2\pi \frac{x-2}{x^2-1} \delta x$$

$$= 2\pi \int_2^4 \frac{x-2}{x^2-1} dx$$

ii) $V = \pi \int_2^4 \frac{2x}{x^2-1} dx - 4\pi \int_2^4 \frac{1}{(x-1)(x+1)} dx,$

$$\frac{1}{(x-1)(x+1)} = \frac{A}{x-1} + \frac{B}{x+1}$$

$$1 = A(x+1) + B(x-1) \Rightarrow \begin{aligned} x=1, & A = \frac{1}{2} \\ x=-1, & B = -\frac{1}{2} \end{aligned}$$

$$\therefore V = \pi \int_2^4 \frac{2x}{x^2-1} dx - \frac{4\pi}{2} \int_2^4 \left(\frac{1}{x-1} - \frac{1}{x+1} \right) dx$$

$$= \pi \left[\ln(x^2-1) \right]_2^4 - 2\pi \left[\ln(x-1) - \ln(x+1) \right]_2^4$$

$$= \pi (\ln 15 - \ln 3) - 2\pi (\ln 3 - \ln 5) - (\ln 1 - \ln 3)$$

$$= \pi \ln 5 - 2\pi (2 \ln 3 - \ln 5)$$

$$= \pi \left(\ln 5 - \ln \frac{9^2}{5^2} \right)$$

$$= \pi \left(\ln \frac{5^3}{3^4} \right)$$

$$= \pi \ln \frac{125}{81}$$

b) $a > 0, b > 0, c > 0$ $a+b+c=1$

$$(x-y)^2 \geq 0$$

$$x^2 - 2xy + y^2 \geq 0$$

$$\therefore x^2 + y^2 \geq 2xy$$

let $x = \sqrt{a}$ and $y = \sqrt{b}$

so $a+b \geq 2\sqrt{ab}$ -i

Similarly, $b+c \geq 2\sqrt{bc}$ -ii

$c+a \geq 2\sqrt{ca}$ -iii

1x11x111

$$(a+b)(b+c)(c+a) \geq 2\sqrt[3]{abbcac}$$

$$\geq 8\sqrt[3]{a^2b^2c^2} \geq 8abc$$

ii)

$$a+b+c=1$$

$$\text{so } a+b=1-c$$

$$b+c=1-a$$

$$c+a=1-b$$

$$\text{so } (1-a)(1-b)(1-c) = (b+c)(c+a)(a+b) \\ \geq 8abc.$$

e). i).

$$\text{LHS} = \int_0^a F(x) dx$$

$$= \left[f(x) \right]_0^a$$

$$= f(a) - f(0)$$

$$\text{where } \frac{d}{dx} f(x) = F(x)$$

$$\text{RHS} = \int_0^a F(a-x) dx$$

$$= \left[(-1)f(a-x) \right]_0^a$$

$$= -f(a-a) - (-f(a-0))$$

$$= -f(0) + f(a)$$

$$= \text{LHS}$$

ii)

$$I = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \log_e(1 + \tan \theta) d\theta$$

$$= \int_0^{\frac{\pi}{4}} \log_e(1 + \tan(\frac{\pi}{4} - \theta)) d\theta \quad \text{using (i)}$$

$$= \int_0^{\frac{\pi}{4}} \log_e \left(1 + \frac{\tan \frac{\pi}{4} - \tan \theta}{1 + \tan \frac{\pi}{4} \cdot \tan \theta} \right) d\theta.$$

$$= \int_0^{\frac{\pi}{4}} \log_e \left(1 + \frac{1 - \tan \theta}{1 + \tan \theta} \right) d\theta$$

$$= \int_0^{\frac{\pi}{4}} \log_e \left(\frac{2}{1 + \tan \theta} \right) d\theta.$$

$$= \int_0^{\frac{\pi}{4}} \log_e 2 d\theta - \int_0^{\frac{\pi}{4}} (1 + \tan \theta) d\theta.$$

$$= \int_0^{\frac{\pi}{4}} \log_e 2 d\theta - I$$

$$2I = \left[\log_e 2 \times \theta \right]_0^{\frac{\pi}{4}}.$$

$$I = \frac{\pi}{8} \ln 2.$$

Question 16

- a) 12! ways of arranging all books
 3! for each package,
 4! packages

$$\begin{aligned} \text{ways} &= \frac{12!}{(3!)^4 \cdot 4!} \\ &= 15400 \end{aligned}$$

- b) each trapezia width is $\frac{n-1}{n-1} = 1$ — n function calc

$$\begin{aligned} \int_1^n \ln x dx &= \frac{1}{2} \left[\ln(1) + \ln(n) + 2(\ln 2 + \ln 3 + \ln 4 + \dots + \ln(n-1)) \right] \\ &= \frac{1}{2} \left[0 + \ln(n) + 2 \ln(2 \cdot 3 \cdot 4 \cdot \dots \cdot (n-1)) \right] \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2} \left[\ln(n!) 2 \ln(n-1)! \right] \\
&= \frac{1}{2} \left[2 \ln(n!) - \ln(n) \right] \\
&= \ln(n!) - \frac{1}{2} \ln(n)
\end{aligned}$$

ii) $\frac{d}{dx}(x \ln x) = \cancel{x} \cdot \frac{1}{\cancel{x}} + 1 \cdot \ln x - 1$
 $= \ln x.$

$$\begin{aligned}
\int_1^n \ln x \, dx &= \left[x(\ln x - 1) \right]_1^n \\
&= n \ln(n) - n - (1 \cdot \ln(1) - 1) \\
&= n \ln(n) - n + 1
\end{aligned}$$

iii) as $y = \ln x$ is concave down,
the sum of areas of trapezia will
be less than the exact area under
the curve.

Hence

$$\ln(n!) - \frac{1}{2} \ln(n) < n \ln(n) - n + 1$$

$$\begin{aligned}
\ln(n!) &< \frac{1}{2} \ln(n) + n \ln(n) - n + 1 \\
&< \left(n + \frac{1}{2}\right) \ln(n) - n + 1
\end{aligned}$$

$$c) i) I_n = \int_1^e (1 - \ln x)^n dx.$$

$$\text{let } u = (1 - \ln x)^n, \quad du = n(1 - \ln x)^{n-1} \cdot \left(-\frac{1}{x}\right) dx$$

$$dv = dx \quad \cancel{x} = x$$

$$\begin{aligned} I_n &= \left[x(1 - \ln x) \right]_1^e - \int_1^e x \cdot n(1 - \ln x)^{n-1} \cdot \left(-\frac{1}{x}\right) dx \\ &= \left(e(1 - \ln e) - 1(1 - \ln 1) \right) + n \int_1^e (1 - \ln x)^{n-1} dx \\ &= e(1-1) - 1(1-0) + n I_{n-1} \\ &= -1 + n I_{n-1} \end{aligned}$$

$$ii) I_3 = -1 + 3I_2.$$

$$I_2 = -1 + 2I_1$$

$$I_1 = -1 + I_0$$

$$\begin{aligned} I_0 &= \int_1^e (1 - \ln x)^0 dx \\ &= \left[x \right]_1^e \\ &= e - 1 \end{aligned}$$

$$\begin{aligned} I_1 &= -1 + e - 1 \\ &= e - 2 \end{aligned}$$

$$\begin{aligned} I_2 &= -1 + 2(e - 2) \\ &= 2e - 5 \end{aligned}$$

$$\begin{aligned} I_3 &= -1 + 3(2e - 5) \\ &= 6e - 16 \end{aligned}$$

iii) Show $\frac{I_n}{n!} = e - \sum_{r=0}^n \frac{1}{r!}$, $n=1, 2, 3, \dots$

Prove true for $n=1$

$$\text{LHS} = \frac{I_1}{1!}$$

$$= \frac{e-2}{1}$$

$$= e-2$$

$$\text{RHS} = e - \sum_{r=0}^1 \frac{1}{r!}$$

$$= e - \left(\frac{1}{0!} + \frac{1}{1!} \right)$$

$$= e - \left(1 + 1 \right)$$

$$= e-2$$

$$= \text{LHS}$$

$\therefore$ true for $n=1$

Assume true for $n=k$.

$$\therefore \frac{I_k}{k!} = e - \sum_{r=0}^k \frac{1}{r!}$$

Prove true for $n=k+1$

$$\text{i.e. } \frac{I_{k+1}}{(k+1)!} = e - \sum_{r=0}^{k+1} \frac{1}{r!}$$

$$\text{LHS} = \frac{I_{k+1}}{(k+1)!}$$

$$= \frac{-1 + (k+1)I_k}{(k+1)!}$$

$$= \frac{-1}{(k+1)!} + \frac{I_k}{k!}$$

$$= \frac{-1}{(k+1)!} + e - \sum_{r=0}^k \frac{1}{r!}$$

$$= e - \left(\sum_{r=0}^k \frac{1}{r!} + \frac{1}{(k+1)!} \right)$$

$$= e - \sum_{r=0}^{k+1} \frac{1}{r!} = \text{RHS}$$

$\therefore$ by induction,
statement is true.